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Geometric Sequence / PROGRESSION (G.P)

$a, ar, ar^2, ar^3, \ldots, ar^{n-1}$

$a_n = ar^{n-1}$ (General / n^{th} term)

Here

$a_n = n^{th}$ term

$a =$ first term

$r =$ common ratio

Ex # 4.5

Q1 Find the indicated term:-

i) $3, 6, 12, \ldots, 5^{th}$ term.

Here,

$a = 3, \quad r = \frac{6}{3} = 2, \quad n = 5$

$\therefore a_n = ar^{n-1}$

$a_5 = ar^{5-1}$

$a_5 = (3)(2)^4$

$a_5 = 48$

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iv) $i, 1-i, \dots; \quad a_{16} = ?$

Here,

$a = i$

$r = \frac{1-i}{i}$

$r = \frac{1-i}{i} \times \frac{-i}{-i}$

$r = \frac{-i+i^2}{-i^2}$

$r = \frac{-i-1}{-(-1)}$

$\boxed{r = -1-i}$

$\therefore \quad a_n = ar^{n-1}$

$a_{16} = (i)(-1-i)^{16-1}$

$a_{16} = i(-1-i)^{15}$

$a_{16} = (i)(-1-i)(-1-i)^{14}$

$a_{16} = (-i-i^2)\{(-1-i)^2\}^7$

$a_{16} = (-i+1)\{(-1)^2-2(-1)(i)+(i)^2\}^7$

$a_{16} = (1-i)(\cancel{1}+2i-\cancel{1})^7$

$a_{16} = (1-i)(2i)^7$

$a_{16} = (1-i)(2)^7(i)^7$

$a_{16} = (1-i)(128)(i)(i)^6$

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$a_{16} = (1-i)(128i)(i^2)^3$

$a_{16} = (1-i)(128i)(-1)^3$

$a_{16} = (128i - 128i^2)(-1)$

$a_{16} = -128i + 128i^2$

$a_{16} = -128i - 128$

Q2 $a_1 = \frac{5}{9}$, $a_6 = \frac{15625}{9}$, $a_n = ?$

$\therefore a_n = ar^{n-1}$

put $n = 6$

$a_6 = ar^{6-1}$

$\frac{15625}{\cancel{9}} = \left(\frac{5}{\cancel{9}}\right) r^5$

$\frac{15625}{5} = r^5$

$\sqrt[5]{15625} = r$

$\boxed{r = 5}$

$\therefore a_n = ar^{n-1}$

$= \left(\frac{5}{9}\right)(5)^{n-1} \Rightarrow a_n = \frac{(5)^n}{9}$

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Q1)

v) $b^2-c^2,\ b+c,\ \dots,$ 9th term?

Here,

$$a = b^2 - c^2 = (b+c)(b-c)$$

$$r = \frac{\cancel{b+c}}{\cancel{(b+c)}(b-c)}$$

$$r = \frac{1}{b-c}$$

$$a_n = ar^{n-1}$$

$$a_9 = (b+c)(b-c)\left(\frac{1}{b-c}\right)^{9-1}$$

$$a_9 = (b+c)(b-c)\left(\frac{1}{b-c}\right)^{8}$$

$$a_9 = \frac{(b+c)\cancel{(b-c)}}{(b-c)^8}$$

$$a_9 = \frac{(b+c)}{(b-c)^7}$$

Q) Find the nth term of the geometric sequence if

$\frac{a_5}{a_3} = \frac{4}{9} \rightarrow (i)$ and $a_2 = \frac{4}{9}$

$a_n = ?$, $a = ?$, $r = ?$, $n = n$

$\therefore a_n = ar^{n-1}$

$a_2 = ar^{2-1}$

$\frac{4}{9} = ar$

$a = \frac{4}{9r} \rightarrow (ii)$

for r,

$a_n = ar^{n-1}$

$a_5 = ar^{5-1}$

$a_5 = ar^4$

for a_3,

$a_3 = ar^{3-1}$

$a_3 = ar^2$

$\therefore \frac{a_5}{a_3} = \frac{4}{9}$

put in eq (i)

$$\frac{a\cancel{r}^4}{a\cancel{r^2}} = \frac{4}{9}$$

$$r^2 = \frac{4}{9}$$

$$r = \pm\sqrt{\frac{4}{9}}$$

$$\boxed{r = \frac{\pm 2}{3}}$$

when $r = \frac{2}{3}$

$$a = \frac{4}{9r}$$

$$a = \frac{\cancel{4}\,2}{{}^{3}\cancel{9}\left(\frac{\cancel{2}}{\cancel{3}}\right)}$$

$$a = \frac{2}{3}$$

$$\therefore a_n = ar^{n-1}$$

$$a_n = \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)^{n-1}$$

$$a_n = \left(\frac{2}{3}\right)^n$$

~~$\therefore a_n =$~~

When $r = -\frac{2}{3}$

$$a = \frac{4}{9\left(-\frac{2}{3}\right)}$$

$$a = \frac{-2}{3}$$

$$a_n = \left(\frac{-2}{3}\right)\left(\frac{-2}{3}\right)^{n-1}$$

$$a_n = \left(\frac{-2}{3}\right)^{n}$$

Q4 Show that the reciprocal of the terms of the G.P. $a_1, a_1r^2, a_1r^4, \ldots$ form another G.P.

Sol:-

$$r = \frac{a_2}{a_1} \quad ; \quad r = \frac{a_3}{a_2}$$

$$r = \frac{\cancel{a_1}r^2}{\cancel{a_1}} \quad ; \quad r = \frac{\cancel{a_1r^4}}{\cancel{a_1r^2}}$$

$$\boxed{r = r^2} \quad ; \quad \boxed{r = r^2}$$

now for reciprocals,

$$\frac{1}{a_1}, \frac{1}{a_1r^2}, \frac{1}{a_1r^4}, \ldots$$

$$r = \frac{\frac{1}{\cancel{a_1}r^2}}{\frac{1}{\cancel{a_1}}} \quad ; \quad r = \frac{\frac{1}{\cancel{a_1r^4}}}{\frac{1}{\cancel{a_1r^2}}}$$

$$\boxed{r = \frac{1}{r^2}} \; ; \; \boxed{r = \frac{1}{r^2}}$$

Hence proved!

Q5 How many terms are in G.P

$$\frac{1}{3}, \frac{1}{12}, \frac{1}{48}, \ldots, \frac{1}{196608}$$

∴ formula of GP:

$$a_n = ar^{n-1} \rightarrow (i)$$

$$a_n = \frac{1}{196608}, \quad a = \frac{1}{3}, \quad r = \frac{\frac{1}{12}}{\frac{1}{3}} = \frac{1}{4}$$

put in eq (i)

$$\frac{1}{196608} = \left(\frac{1}{3}\right)\left(\frac{1}{4}\right)^{n-1}$$

$$\frac{3}{196608} = \left(\frac{1}{4}\right)^{n-1}$$

$$\frac{1}{65536} = \left(\frac{1}{4}\right)^{n-1}$$

$$\cancel{\left(\frac{1}{4}\right)}^{8} = \cancel{\left(\frac{1}{4}\right)}^{n-1}$$

$$8 = n-1$$

$$\boxed{9 = n}$$

Q. Find the three consecutive numbers whose sum is 39 and their product is 729.

Sol:-

$$\frac{a}{r},\ a,\ ar$$

for product

$$\left(\frac{a}{\cancel{r}}\right)(a)(a\cancel{r}) = 729$$

$$a^3 = 729$$

$$a = \sqrt[3]{729}$$

$$\boxed{a = 9}$$

for sum,

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$$\frac{9}{r} + 9 + 9r = 39$$

$$\frac{9 + 9r + 9r^2}{r} = 39$$

$$9(1 + r + r^2) = 39r$$

$$\cancel{9}^{3}(1 + r + r^2) = \cancel{39}^{13}r$$

$$3 + 3r + 3r^2 = 13r$$

$$3 + 3r - 13r + 3r^2 = 0$$

$$3r^2 - 10r + 3 = 0$$

$$3r^2 - 9r - r + 3 = 0$$

$$3r(r-3) - 1(r-3) = 0$$

$$(r-3)(3r-1) = 0$$

$r - 3 = 0$	$3r - 1 = 0$
$r = 3$	$3r = 1$
	$r = \frac{1}{3}$

for $r = 3$ and $a = 9$

$$\frac{9}{3},\ 9,\ 9(3)$$

$3,\ 9,\ 27$

for $r = \frac{1}{3}$ and $a = 9$

$\frac{9}{\left(\frac{1}{3}\right)}, 9, \frac{9^3}{} \times \frac{1}{3}$

27, 9, 3

Q7 The number of bacteria in a culture increased in G.P from 515,000 to 15,45,000 in 7 days. Find the daily rate of increase, assuming the rate of increase to be constant.

$r = ?$

$a_1 = a = 515,000$

$n = 1 + 7 = 8$

$r = \frac{a_2}{a_1}$

$\because a_n = ar^{n-1}$

$a_8 = ar^{8-1}$

$a_8 = ar^7$

$1545000 = 515000\, r^7$

$r^7 = \frac{15,45,000}{515,000}$

$r = \sqrt[7]{3}$

$r = 1.169$

$r = 1.17 \simeq$

Q8 Find profit on Rs 1000 for 5 years at 4% per annum compound profit.

$a = 1000, \quad r = \frac{4}{100} = 0.04 + 1$

$r = 1.04$

$n = 5 + 1 = 6$

$a_6 = (1000)(1.04)^{6-1}$

$a_6 = (1000)(1.04)^5$

$a_6 = 1216.65$

$a_6 = 1217$

Profit = total - investment

Profit = 1217 - 1000

= 217 Rs

(or)

$\text{Profit} = (1000)(1.04)^5 - 1000$

$= 1000\{(1.04)^5 - 1\}$